

A COMPARISON OF NEW AND EXISTING BAYESIAN TWO-STAGE DESIGNS

Arvind Ruggoo

Faculty of Agriculture, University of Mauritius, Reduit, Mauritius

and

Martina Vandebroek

Faculty of Business and Economics, Leuven Statistics Research Centre,
Katholieke Universiteit Leuven, Naamsestraat 69, B3000 Leuven, Belgium

Key words: Bayesian $\mathcal{D}$ -optimal design, Bayesian $\mathcal{D}$ - $\mathcal{D}$ optimal design, two-stage designs, model-robust designs, posterior probabilities, bias, lack-of-fit, follow-up designs.

Summary: In this paper we revisit the work of DuMouchel and Jones (1994) and combine their Bayesian $\mathcal{D}$ -optimality procedure with the Bayesian update procedure of Jones and DuMouchel (1996) in a two-stage design strategy. The resulting Bayesian $\mathcal{D}$ - $\mathcal{MF}$ procedure is robust against departures from an assumed model whilst allowing good fit of the assumed model. The procedure is then compared to existing designs and recommendations are made on their implementation.

1. Introduction

It is common knowledge that if we use a simple linear function to estimate the expected values of the response when the true model is quadratic, it will result in a large bias term. Therefore in the literature, while designing experiments for regression models, model-robustness has always been an important issue. Chang and Notz (1996) provide a comprehensive review on the subject. According to them, a good model robust design should:

1. Allow one to fit the assumed model.

AMS: 62K05

2. Detect the model inadequacy when the fitted model is a poor approximation to the true model.
3. Allow one to make reasonable efficient inferences concerning the assumed model when the assumed model is adequate.

Considerable progress has been made in this area and our intention here is not to provide a reference list on the subject (for this, see papers like Wiens (1994), Yue and Hickernell (1999), Heo, Schmuland and Wiens (2001), Fang and Wiens (2003) and references therein). Model uncertainty also obviates the use of standard optimal design procedures which do not offer the possibility to distinguish between primary and potential terms in the response model. The penalty for excluding necessary explanatory terms will introduce bias in the estimated coefficients. However, we note that most of the work in this area has considered the class of approximate designs and has not necessarily led to practical designs for the experimenter. For exact designs and from a practical point of view, the very interesting work of DuMouchel and Jones (1994) illustrates the use of Bayesian methods for the selection of a design in which model uncertainty does not allow the use of a $\mathcal{D}$ -optimal design. They assume two types of model terms: primary and potential terms. By conceiving a prior distribution on the model coefficients they produce a practical way of locating a Bayesian $\mathcal{D}$ -optimal design that maximizes the determinant of the posterior information matrix.

Andere-Rendon, Montgomery and Rollier (1997) make use of this idea in developing experiments involving mixtures. Neff (1996) demonstrates the advantage of the DuMouchel and Jones (1994) method and illustrates a two-stage Bayesian $\mathcal{D}$ - $\mathcal{D}$ optimal approach in which prior information is updated at the end of the first stage. Ruggoo and Vandebroek (2004) have reviewed

and extended the work of Neff (1996) by proposing some modifications in her algorithm to handle uncertainty in the prior distribution specifications.

In a similar vein for screening experiments, Meyer, Steinberg and Box (1996) have proposed within the Bayesian paradigm a model-discrimination ($\mathcal{MD}$) criterion for designing follow-up experiments when results of a fractional experiment are ambiguous due to confounding among the possible effects, and more than one model may be consistent with the data. In the discussion following that article, Jones and DuMouchel (1996) propose an extension of their original work of 1994, and develop an optimization criterion for design augmentation that solves the same problem as Meyer, Steinberg and Box (1996) for adding extra runs in screening experiments. The $\mathcal{MF}$ -criterion they propose, is simpler and computationally much faster than the $\mathcal{MD}$ -criterion and is applicable to the same problem setting. (Jones and DuMouchel (1996) used the letter $\mathcal{F}$ for the criterion to enforce the analogy to the Fisher's information matrix. We use the notation $\mathcal{MF}$ to avoid confusion with the $\mathcal{F}$ -optimal design that minimizes the length of the Fieller interval for the ED50 in logistic models (see Letsinger, 1995).) In essence, the $\mathcal{MF}$ -criterion is similar to Bayesian $\mathcal{D}$ -optimality of DuMouchel and Jones (1994), but making explicit use of the actual results of an initial experiment.

The procedure of DuMouchel and Jones (1994) generates designs that are more robust to the bias caused by an incorrect model. Their approach preserves the flexibility and ease of use of the algorithmic designs and is available in the OPTEx procedure of SAS. The $\mathcal{MF}$ -criterion of Jones and DuMouchel (1996) has been developed within the same Bayesian framework and intuitively it would be interesting to combine the original work of DuMouchel and Jones (1994) and use the $\mathcal{MF}$ -criterion of Jones and DuMouchel (1996) for empirical models in a two-stage design strategy. This would entail using the

approach of DuMouchel and Jones (1994) in the first stage to get a design with reduced dependence on the assumed model. The $\mathcal{MF}$ -criterion makes use of first stage data and can be used to generate the second stage design. This combined first and second stage design should possess attractive properties in terms of precision of the parameters of the assumed model, lack-of-fit and bias in the direction of the potential terms. We shall also compare this extension to other existing procedures.

In Section 2, we describe and adapt the $\mathcal{MF}$ -criterion of Jones and DuMouchel (1996) developed for follow-up trials in screening experiments in a two-stage strategy by combining it with the Bayesian $\mathcal{D}$ -optimal procedure of DuMouchel and Jones (1994). In Section 4, we illustrate and compare this approach to existing one-stage and two-stage procedures. Finally Section 5 contains a small discussion and some practical recommendations for any experimenter wishing to use a two-stage procedure.

2. Development of the Bayesian $\mathcal{D}$ - $\mathcal{MF}$ procedure

Suppose the linear model that will be fitted by the experimenter is of the form

$$y = \mathbf{x}'_{pri} \boldsymbol{\beta}_{pri} + \varepsilon,$$

with $\mathbf{x}_{pri}$ being a p -dimensional vector of powers and products of the experimental factors and $\boldsymbol{\beta}_{pri}$ the p -dimensional vector of unknown parameters attached to the primary terms. The product $\mathbf{x}'_{pot} \boldsymbol{\beta}_{pot}$ contains the terms that one wishes to protect against in designing the experiment, where $\mathbf{x}_{pot}$ is the q -dimensional vector containing powers and products of the factors and $\boldsymbol{\beta}_{pot}$ is the q -dimensional vector associated with the potential terms. The model is reparameterized in terms of the orthonormal polynomials with respect to a measure μ on the design region which ensures that the effects are well separable and independent so that a simple prior distribution on the potential

terms can be used. Therefore, it becomes sensible to allocate an uncertainty prior for the potential effects that incorporates common high variance but independent variables with zero mean. The prior distribution of β_{pot} is thus assumed to be $N(\mathbf{0}, \tau^2 \sigma^2 \mathbf{I}_q)$ where τ^2 is the common prior variance of the potential terms' coefficients, measured in units of the random error variance σ^2 of the error terms. Also $\mathbf{y}_i | \beta \sim N(\mathbf{X}_i \beta, \sigma^2 \mathbf{I}_{n_i})$ for each stage i ($i = 1, 2$) and the first and second stage comprise n_1 and n_2 runs respectively so that the total number of design points in the combined design is $n_1 + n_2$. $\mathbf{X}$ is the extended design matrix of dimension $n \times (p + q)$ for the combined stages, so that $\mathbf{X}' = [\mathbf{X}'_1 \ \mathbf{X}'_2]$. $\mathbf{X}_1 = [\mathbf{X}_{pri(1)} \ \mathbf{X}_{pot(1)}]$ is of dimension $n_1 \times (p + q)$ and $\mathbf{X}_2 = [\mathbf{X}_{pri(2)} \ \mathbf{X}_{pot(2)}]$ is of dimension $n_2 \times (p + q)$. These matrices represent respectively the first and second stage designs expanded to full model space. $\mathbf{X}_{pri(i)}$ and $\mathbf{X}_{pot(i)}$ correspond to the primary and potential terms respectively for each stage i ($i = 1, 2$). Finally $\mathbf{X}'_{pri} = [\mathbf{X}'_{pri(1)} \ \mathbf{X}'_{pri(2)}]$ is of dimension $n \times p$ and $\mathbf{X}'_{pot} = [\mathbf{X}'_{pot(1)} \ \mathbf{X}'_{pot(2)}]$ is of dimension $n \times q$. These matrices are respectively the combined first and second stage design matrices for the primary and potential terms models only.

The first stage design is obtained using the approach of DuMouchel and Jones (1994) by numerically maximizing

$$|\mathbf{X}'_1 \mathbf{X}_1 + \mathbf{K} / \tau^2| \quad (1)$$

with respect to $\mathbf{X}_1$ and where $\mathbf{K}$ is a $(p + q) \times (p + q)$ diagonal matrix, whose first p diagonal elements are equal to 0 and the remaining q diagonal elements are equal to 1. Before observing the first stage data, the experimenter has a model with $(p + q)$ regressors. Consequently each candidate model M_i contains all primary terms and a subset of q_i ($0 \leq q_i \leq q$) potential terms so that the total number of possible models is $m = 2^q$. Now at the end of the first stage,

the posterior mean and variance of β_i given model M_i are

$$\hat{\beta}_i = E(\beta_i | M_i, \mathbf{y}_1) = \left(\mathbf{X}'_{1(i)} \mathbf{X}_{1(i)} + \frac{\mathbf{K}_i}{\tau^2} \right)^{-1} \mathbf{X}'_{1(i)} \mathbf{y}_1 \quad (2)$$

and

$$\mathbf{V}_i = \sigma^2 \left(\mathbf{X}'_{1(i)} \mathbf{X}_{1(i)} + \frac{\mathbf{K}_i}{\tau^2} \right)^{-1} \quad (3)$$

where $\mathbf{X}_{1(i)}$ is the first stage design in model M_i space and of dimension $n_1 \times (p + q_i)$. Using the approach of Box and Meyer (1993), the resulting posterior probability for model M_i becomes

$$\begin{aligned} p(M_i | \mathbf{y}_1) &= C \pi^{q_i} (1 - \pi)^{q - q_i} \tau^{-q_i} \left| \mathbf{X}'_{1(i)} \mathbf{X}_{1(i)} + \frac{\mathbf{K}_i}{\tau^2} \right|^{-1/2} \\ &\quad \times (\mathbf{S}(\hat{\beta}_i) + \frac{1}{\tau^2} \hat{\beta}_i' \mathbf{K}_i \hat{\beta}_i)^{-(n_1 - 1)/2}, \end{aligned} \quad (4)$$

where π is the prior probability that any one of the potential terms is active in the true model, $\mathbf{S}(\hat{\beta}_i)$ is the residual sum of squares for model M_i and C is the normalization constant (see Ruggoo & Vandebroek, 2004 for additional details).

We now use all this posterior information for all the $m = 2^q$ models from the first stage experiment as prior information for our future experimental observations. Consequently the prior expected coefficient vector for the second stage design is

$$\beta_0 = \sum_{M_i} p(M_i | \mathbf{y}_1) \hat{\beta}_i^*.$$

β_0 can be thought of as the weighted average of the posterior mean for the set of m sub-models. $\hat{\beta}_i^*$ is a vector that comes from elements of (2). It consists of as many elements as the number of unique terms in all the models. Consequently it is a vector of dimension $(p + q)$ which corresponds to the full model with all primary and potential terms. If a term is not in model M_i , then $\hat{\beta}_i^*$ for that term is 0 (Jones and DuMouchel, 1996).

To illustrate the construction of the vector $\hat{\beta}_i^*$, consider the simple example where $\mathbf{x}^{(\text{pri})} = \{1, x_1, x_2\}$ and $\mathbf{x}^{(\text{pot})} = \{x_1x_2, x_1^2\}$. The $m = 4$ possible models are:

Model I	1	x_1	x_2	
Model II	1	x_1	x_2	x_1x_2
Model III	1	x_1	x_2	x_1^2
Model IV	1	x_1	x_2	x_1x_2 x_1^2

Here $p = 3$ and $q = 2$ and consequently the vectors $\hat{\beta}_i^*$, ($i = 1, \dots, 4$) will be column vectors with five elements each. If we fit all the models above, the vectors $\hat{\beta}_i^*$ are respectively

$$\begin{aligned}\hat{\beta}_1^* &= [\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, 0, 0]' \\ \hat{\beta}_2^* &= [\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_{12}, 0]' \\ \hat{\beta}_3^* &= [\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, 0, \hat{\beta}_{11}]' \\ \hat{\beta}_4^* &= [\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_{12}, \hat{\beta}_{11}]'. \end{aligned}$$

Now the prior covariance matrix for the second stage design is

$$\mathbf{V}_0 = \sum_{M_i} p(M_i|\mathbf{y}_1) \mathbf{V}_i^* + \sum_{M_i} p(M_i|\mathbf{y}_1) (\hat{\beta}_i^* - \beta_0)(\hat{\beta}_i^* - \beta_0)'. \quad (5)$$

Note that this expression has specifically been proposed by Jones and DuMouchel (1996) in this context. It can be compared to the mean squared error term, which is the sum of the variance and the squared bias. The prior variance needs to incorporate uncertainty about the mean model. The first term on the right-hand side of (5) is the weighted average of variance-covariance matrices from the set of sub-models being considered. $\mathbf{V}_i^*$ is the square matrix with as many rows and columns as there are unique terms and their components emanate from elements of matrix (3). Consequently, the dimension of $\mathbf{V}_i^*$ is $(p + q) \times (p + q)$ and the entries that correspond to terms not in model M_i

are 0. The second term on the right-hand side of (5) is the weighted average of the outer product of the difference between the prior coefficient vector and the individual fitted parameters for each model. This term incorporates the uncertainty about the parameters into the prior variance for augmentation. Note that in the Bayesian paradigm, the model parameters are not fixed unknown parameters but they are random variables. What a frequentist would think of as bias can be incorporated as variance when considering a set of possible models (see Gelman et al. (2003)). Draper and Guttman (1992) also treat the bias as a variance for experimental design purposes by using a variance function as a criterion and incorporating the bias errors as random rather than fixed effects. So the second term on the right-hand side of (5) is the bias term that incorporates the uncertainty about the mean model.

To illustrate computation of the matrices, $\mathbf{V}_i^*$, we continue with the same example with the four possible models. For the full model, we have

$$\mathbf{V}_4 = \begin{bmatrix} V_{11} & V_{12} & V_{13} & V_{14} & V_{15} \\ V_{21} & V_{22} & V_{23} & V_{24} & V_{25} \\ V_{31} & V_{32} & V_{33} & V_{34} & V_{35} \\ V_{41} & V_{42} & V_{43} & V_{44} & V_{45} \\ V_{51} & V_{52} & V_{53} & V_{54} & V_{55} \end{bmatrix}_{5 \times 5}.$$

The matrices $\mathbf{V}_i^*$ ($i = 1, \dots, 4$) will also be of dimensions 5×5 and the entries that correspond to terms not in model M_i are 0 as below:

$$\mathbf{V}_1^* = \begin{bmatrix} V_{11} & V_{12} & V_{13} & 0 & 0 \\ V_{21} & V_{22} & V_{23} & 0 & 0 \\ V_{31} & V_{32} & V_{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{5 \times 5},$$

$$\begin{aligned}
\mathbf{V}_2^* &= \begin{bmatrix} V_{11} & V_{12} & V_{13} & V_{14} & 0 \\ V_{21} & V_{22} & V_{23} & V_{24} & 0 \\ V_{31} & V_{32} & V_{33} & V_{34} & 0 \\ V_{41} & V_{42} & V_{43} & V_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}^{5 \times 5}, \\
\mathbf{V}_3^* &= \begin{bmatrix} V_{11} & V_{12} & V_{13} & 0 & V_{15} \\ V_{21} & V_{22} & V_{23} & 0 & V_{25} \\ V_{31} & V_{32} & V_{33} & 0 & V_{35} \\ 0 & 0 & 0 & 0 & 0 \\ V_{51} & V_{52} & V_{53} & 0 & V_{55} \end{bmatrix}^{5 \times 5}, \\
\mathbf{V}_4^* &= \begin{bmatrix} V_{11} & V_{12} & V_{13} & V_{14} & V_{15} \\ V_{21} & V_{22} & V_{23} & V_{24} & V_{25} \\ V_{31} & V_{32} & V_{33} & V_{34} & V_{35} \\ V_{41} & V_{42} & V_{43} & V_{44} & V_{45} \\ V_{51} & V_{52} & V_{53} & V_{54} & V_{55} \end{bmatrix}^{5 \times 5}.
\end{aligned}$$

Jones and DuMouchel (1996) suggest the following as an optimization criterion for design augmentation in the context of follow-up experiments for screening trials:

$$\mathcal{MF} = |\mathbf{Z}'\mathbf{Z} + \mathbf{V}_0^{-1}| |\mathbf{V}_0|$$

where $\mathbf{Z}$ is the design matrix of new runs using the full model (the union of terms in the individual m models). Scaling by the determinant of $\mathbf{V}_0$ does not change the optimal design, but allows to view $\mathcal{MF}$ as a measure of incremental information supplied by the extra runs (Jones and DuMouchel, 1996).

Therefore, applying the same line of thought in a two-stage design strategy, suggests that our second-stage design can be obtained by maximizing

$$\mathcal{MF} = |\mathbf{X}_2'\mathbf{X}_2 + \mathbf{V}_0^{-1}| \quad (6)$$

where $\mathbf{X}_2$ is the extended design matrix of new runs in the second stage using the full model. We note that the $\mathcal{MF}$ -criterion bears similarity to criterion

(1) of DuMouchel and Jones (1994) with prior information on the parameters updated using model information from first stage data. In essence the criterion allows the effect of a first stage data to emerge through Bayesian upgrading both of the posterior distribution for the coefficient terms given the model and first stage data, and of the posterior probability for the appropriate model given the first stage data vector. We shall refer to this new two-stage procedure as the Bayesian $\mathcal{D}\text{-}\mathcal{MF}$ procedure.

Briefly, the Bayesian $\mathcal{D}\text{-}\mathcal{D}$ approach of Ruggoo and Vandebroek (2004) is as follows: the Bayesian $\mathcal{D}$ -optimality criterion of DuMouchel and Jones (1994) which maximizes $|\mathbf{X}'_1 \mathbf{X}_1 + \frac{\mathbf{K}}{\tau^2}|$ is used to select the first stage design. $\tau = 5$ is recommended in the first stage. At the end of the first stage, a convenient update, $\hat{\tau}^2$ of τ^2 is obtained as the mode of the posterior density of $f(\tau^2|\mathbf{y}_1)$. The posterior variance of β_i in the second stage is now

$$\mathbf{V}_{2(i)} = \sigma^2 \left(\mathbf{X}'_{1(i)} \mathbf{X}_{1(i)} + \mathbf{X}'_{2(i)} \mathbf{X}_{2(i)} + \frac{\mathbf{K}_i}{\hat{\tau}^2} \right)^{-1},$$

where $\mathbf{X}_{1(i)}$ and $\mathbf{X}_{2(i)}$ are the first stage and second stage design matrices respectively expanded to model space M_i . A Bayesian $\mathcal{D}\text{-}\mathcal{D}$ design for model M_i is the set of design points $\mathbf{X}_{2(i)}$ which minimizes $|\mathbf{V}_{2(i)}|$. Using the posterior model probabilities computed earlier, the second stage design points $\mathbf{X}_2$ are obtained by minimising

$$\sum_{M_i} p(M_i|\mathbf{y}_1) |\mathbf{V}_{2(i)}|.$$

Ruggoo and Vandebroek (2004) and Neff and Myers (1998) suggest the use of equal sample size partitioning in the two stages, with $n_1 = (p + q + 2)$.

3. Illustration of the Bayesian $\mathcal{D}\text{-}\mathcal{MF}$ procedure

We present in Figure 1, the first and second stage design for one simulation where we have $p = 5$ terms, $\mathbf{x}^{(\text{pri})} = \{1, x_1, x_2, x_3, x_1^2\}$ and $q = 3$ potential

terms, $\mathbf{x}^{(\text{pot})} = \{x_1x_2, x_2^2, x_3^2\}$. Suppose $n_1 = n_2 = (p + q + 2) = 10$ and the true model from which data is simulated consists of the five primary terms and one of the three potential terms:

$$y = 42.0 + 11.5 x_1 + 12.8 x_2 + 10.5 x_3 + 14.6 x_1^2 - 7.4 x_2^2 + \varepsilon. \quad (7)$$

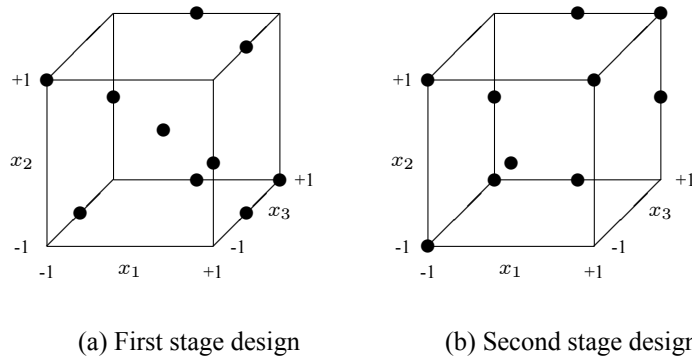


Figure 1 First and second stage designs

The first stage design obtained by the approach of DuMouchel and Jones (1994) has checkpoints that serve as a detection mechanism for the potential terms. The second stage design reflects terms in the true model (7), and has only two levels for x_3 as there is no interest in its quadratic effect. There are several intermediate points on the x_1 and x_2 plane to ensure adequate estimation of the quadratic effects present in the model. Therefore, the second stage design criterion is effective in selecting design points based on information from the first stage design.

4. Comparison of the various two-stage designs

Since the second stage design is dependent on first stage data, we need to evaluate their performance via a simulation approach. The performance of each design will be measured by its efficiency relative to a true assumed model

in the simulations. 200 simulations will be performed and each will produce first stage data for selecting the second stage design. The random error ε , is assumed to be normally distributed with mean 0 and variance 1 and $\pi = 1/3$ in all the simulations. Also $\tau = 1$ is used in the simulations for the Bayesian $\mathcal{D}$ - $\mathcal{MF}$ approach as proposed by DuMouchel and Jones (1994). The competitors to the Bayesian $\mathcal{D}$ - $\mathcal{MF}$ optimal design are the unique stage traditional $\mathcal{D}$ -optimal design for the primary terms model, the Bayesian $\mathcal{D}$ -optimal design of DuMouchel and Jones (1994) and the two-stage Bayesian $\mathcal{D}$ - $\mathcal{D}$ procedure outlined earlier.

The performance of the two-stage procedures will be measured by the average of $D_{X_{pri}}^*$, D_{lof}^* and D_{bias}^* over the different 200 simulations. Actually, $D_{X_{pri}}^* = \left| \mathbf{X}_{pri}^{*'} \mathbf{X}_{pri}^* \right|^{-1/p}$ represents the measure of precision of the primary terms, $D_{lof}^* = \left| \mathbf{L}^* \right|^{-1/q}$ represents a measure of the lack-of-fit component and $D_{bias}^* = \left| \mathbf{A}^{*'} \mathbf{A}^* + \mathbf{I}_q \right|^{1/q}$ represents the degree of bias, where

$$\begin{aligned} \mathbf{L}^* &= \mathbf{X}_{pot}^{*'} \mathbf{X}_{pot}^* - \mathbf{X}_{pot}^{*'} \mathbf{X}_{pri}^* \left(\mathbf{X}_{pri}^{*'} \mathbf{X}_{pri}^* \right)^{-1} \mathbf{X}_{pri}^{*'} \mathbf{X}_{pot}^* \text{ and} \\ \mathbf{A}^* &= \left(\mathbf{X}_{pri}^{*'} \mathbf{X}_{pri}^* \right)^{-1} \mathbf{X}_{pri}^{*'} \mathbf{X}_{pot}^*. \end{aligned}$$

$\mathbf{X}_{pri}^*$ and $\mathbf{X}_{pot}^*$ represent the design points for the primary and potential terms expanded to contain regressors in the true model only. The one-stage $\mathcal{D}$ -optimal design and Bayesian $\mathcal{D}$ -optimal design are not data dependent and can thus be evaluated by the single measures $D_{X_{pri}}^*$, D_{lof}^* and D_{bias}^* for the true model.

We consider the following three cases for our evaluation purposes. The design region considered is the $5 \times 5 \times 5$ grid on $[-1, +1]^3$, whereby the factors are at equally spaced levels: $-1; -0.5; 0; 0.5; 1$. To mimic the effect of imprecise knowledge of a model, the three examples consider the potential versus pseudo-actual quadratic models with all primary terms validly in the

model and particular choices of elements of the corresponding potential terms are allocated non-zero coefficients. This hidden specific model generates data, but all that is known is that the model must involve the primary terms, and may involve one or more of the q potential terms.

Case I:

Suppose we have resources for 20 runs and $n_1 = n_2 = 10$. Let the primary model under consideration for the numerical evaluation be defined with $p = 5$ terms, $\mathbf{x}^{(\text{pri})} = \{1, x_1, x_2, x_3, x_1^2\}$. Suppose also that the researcher wants to protect against the following potential terms, $\mathbf{x}^{(\text{pot})} = \{x_1x_2, x_2^2, x_3^2\}$. For simulation purposes, we shall assume that the true model consists of the five primary terms and one of the three potential terms, so that

$$y = 42.0 + 11.5 x_1 + 12.8 x_2 + 10.5 x_3 + 14.6 x_1^2 - 7.4 x_2^2 + \varepsilon. \quad (8)$$

Case II:

Here $n_1 = n_2 = 11$ and $p = 5$ primary terms, $\mathbf{x}^{(\text{pri})} = \{1, x_1, x_2, x_3, x_1x_2\}$ and $q = 4$ potential terms, $\mathbf{x}^{(\text{pot})} = \{x_1^2, x_1x_3, x_2^2, x_3^2\}$. First stage data is simulated from the true model with the five primary terms and two of the four potential terms as below:

$$y = 42.0 + 11.2 x_1 + 14.5 x_2 + 10.6 x_3 + 12.5 x_1x_2 + 8.9 x_1^2 - 9.9 x_1x_3 + \varepsilon.$$

Case III:

Finally $n_1 = n_2 = 12$ and the model we consider is with $(p + q) = 10$ terms comprising 5 primary terms namely, $\mathbf{x}^{(\text{pri})} = \{1, x_1, x_2, x_3, x_1^2\}$ and 5 potential terms, $\mathbf{x}^{(\text{pot})} = \{x_1x_2, x_1x_3, x_2x_3, x_2^2, x_3^2\}$. For the simulation purposes, the true model has $p = 5$ primary and $q = 3$ potential terms

$$y = 40.0 + 11.5x_1 + 12.8x_2 + 10.5x_3 + 14.6x_1^2 + 9.8x_1x_2 - 7.4x_1x_3 - 8.7x_2^2 + \varepsilon.$$

4.1 Results of the different evaluations

The results of the evaluations are shown in Tables 1–3. Values in brackets are the standard deviations over the 200 simulated data. As expected, the $\mathcal{D}$ -optimal designs have the most desirable precision characteristics but the worst bias. The designs of DuMouchel and Jones (1994) allow to test for lack-of-fit and the small loss in precision is compensated by a drop in the bias in the direction of the potential terms in the true model. Usually when one ‘fears’ model inadequacy there is a need to consider some type of compromise design because a strategy involving protection against bias errors works counter to a strategy designed to minimize variance (see Myers & Montgomery, 2002, for more details). In other words, variance and bias considerations are often conflicting when looking for a model-robust design as a drop in one generally leads to an increase in the other component. The Bayesian $\mathcal{D}\text{-}\mathcal{MF}$ procedure looks promising with designs that have lower bias than their unique stage competitors but still maintaining excellent precision of the assumed primary model. The results are very comparable to the Bayesian $\mathcal{D}\text{-}\mathcal{D}$ procedure of Ruggoo and Vandebroek (2004). The benefit of using previous observations in this new approach is evident with better precision of terms in the primary model than the Bayesian $\mathcal{D}$ -optimal design for the three cases considered. The two-stage approaches seem to be a very interesting compromise which embraces the two conflicting aspects of bias and variance. The resulting designs give a much better reduction in bias compared to the one-stage procedures, whilst still maintaining excellent variance properties for the primary model.

Table 1 Evaluation of the two-stage designs and single-stage competitorsCase I $y = 42.0 + 11.5 x_1 + 12.8 x_2 + 10.5 x_3 + 14.6 x_1^2 - 7.4 x_2^2 + \varepsilon$

Two-stage approach ($n_1 = n_2 = 10$)	Average ($D_{X_{\text{pri}}}^*$)	Average (D_{lof}^*)	Average (D_{bias}^*)
Bayesian $\mathcal{D}\text{-}\mathcal{MF}$ procedure	0.038896 (0.0004634)	0.042475 (0.0031410)	1.204162 (0.0518240)
Bayesian $\mathcal{D}\text{-}\mathcal{D}$ procedure	0.039026 (0.0000892)	0.039208 (0.0004038)	1.115353 (0.0135361)
One-stage approach ($n = 20$)	$D_{X_{\text{pri}}}^*$	D_{lof}^*	D_{bias}^*
$\mathcal{D}$ -optimal (Primary Terms)	0.034299	—	2.428570
DuMouchel and Jones (1994)	0.038914	0.049374	1.279301

Table 2 Evaluation of the two-stage designs and single-stage competitorsCase II $y = 42.0 + 11.2x_1 + 14.5x_2 + 10.6x_3 + 12.5x_1x_2 + 8.9x_1^2 - 9.9x_1x_3 + \varepsilon$

Two-stage approach ($n_1 = n_2 = 11$)	Average ($D_{X_{\text{pri}}}^*$)	Average (D_{lof}^*)	Average (D_{bias}^*)
Bayesian $\mathcal{D}\text{-}\mathcal{MF}$ procedure	0.028645 (0.0011812)	0.027609 (0.0011674)	1.177576 (0.0296951)
Bayesian $\mathcal{D}\text{-}\mathcal{D}$ procedure	0.027327 (0.0001612)	0.028973 (0.0001391)	1.222679 (0.0010328)
One-stage approach ($n = 22$)	$D_{X_{\text{pri}}}^*$	D_{lof}^*	D_{bias}^*
$\mathcal{D}$ -optimal (Primary Terms)	0.022887	—	1.581590
DuMouchel and Jones (1994)	0.02958	0.031216	1.273629

Table 3 Evaluation of the two-stage designs and single-stage competitors

Case III $y =$
 $40.0 + 11.5x_1 + 12.8x_2 + 10.5x_3 + 14.6x_1^2 + 9.8x_1x_2 - 7.4x_1x_3 - 8.7x_2^2 + \varepsilon$

Two-stage approach ($n_1 = n_2 = 12$)	Average ($D_{X_{\text{pri}}}^*$)	Average (D_{lof}^*)	Average (D_{bias}^*)
Bayesian $\mathcal{D}\text{-}\mathcal{MF}$ procedure	0.031514 (0.0006338)	0.021950 (0.0003165)	1.105264 (0.0082149)
Bayesian $\mathcal{D}\text{-}\mathcal{D}$ procedure	0.031708 (0.0002805)	0.021392 (0.0003405)	1.090693 (0.0099708)
One-stage approach ($n = 24$)	$D_{X_{\text{pri}}}^*$	D_{lof}^*	D_{bias}^*
$\mathcal{D}$ -optimal (Primary Terms)	0.028421	—	1.344158
DuMouchel and Jones (1994)	0.031606	0.023785	1.135410

5. Discussion

The main objective of this paper was to develop a two-stage design procedure that combines the Bayesian $\mathcal{D}$ -optimality approach of DuMouchel and Jones (1994) with the Bayesian update procedure of Jones and DuMouchel (1996). The procedure is a natural extension of the Bayesian $\mathcal{D}$ -optimal procedure and uses the simple but powerful idea of incorporating previous observations to guide the choice of new design points. It is worthwhile noting the trade-off associated with moving from the higher precision associated with the estimates for the primary terms when no allowance for potential terms is made, to the proposed modes of taking the path through exploration of the potential terms. An interesting point to stress here is that, as opposed to lot of work devoted to continuous designs and design measures, this work has given a procedure to construct exact two-stage designs which are required in practice and can be implemented by the experimenter. However, it should be noted that the methods presented are only applicable when the assumed primary terms will

be fitted, and hence implicitly assumed to be non-zero. Further work could include the impact of false inclusion of primary terms in the model and the use of more stages in the iteration process.

Acknowledgements

The authors acknowledge support from research funds of the Fund for Scientific Research Flanders. The authors also thank Dr. Bradley Jones for his contribution to this article by clarifying several aspects of the Jones and DuMouchel (1996) criterion. A note of thanks to Prof. Peter Goos for some helpful comments on the paper. Finally the authors would like to thank the editors and in particular the referee for useful clarifications and additions to the text that improved the manuscript.

References

- ANDERE-RENDON, J., MONTGOMERY, D.C. & Rollier, D.A. (1997). Design of mixture experiments using Bayesian D-optimality. *Journal of Quality Technology*, **29**, 451–463.
- BOX, G.E.P. & Meyer, R.D. (1993). Finding the active factors in fractionated screening experiments. *Journal of Quality Technology*, **25**, 94–105.
- CHANG, Y-J. & NOTZ, W.I. (1996). Model robust designs. In: S. Ghosh and C. R. Rao (eds.), *Handbook of statistics* 13, 1055–1098. Elsevier.
- DRAPER, N.R. & Guttman, I. (1992). Treating bias as variance for experimental design purposes. *Annals of the Institute of Mathematical Statistics*, **44(4)**, 659–671.
- DUMOUCHEL, W. & JONES, B. (1994). A simple Bayesian modification of D-optimal designs to reduce dependence on an assumed model. *Technometrics*, **36**, 37–47.
- FANG, Z. & WIENS, D.P. (2003). Robust regression designs for approximate polynomial models. *Journal of Statistical Planning and Inference*, **17**, 305–321.

- GELMAN, A., CARLIN, J.B., STERN, H.S. & RUBIN, D.B. (2003). *Bayesian Data Analysis*. Chapman and Hall/CRC, New York.
- HEO, G., SCHMULAND, B. & WIENS D.P. (2001). Restricted minimax robust designs for misspecified regression model. *The Canadian Journal of Statistics*, **29**(1), 117–128.
- JONES, B. & DUMOUCHEL, W. (1996). Discussion of “Follow-up designs to resolve confounding in fractional factorials”, by R.D. Meyer, D.M. Steinberg & G.E.P. Box. *Technometrics*, **38**, 323–326.
- LETSINGER, W.C. (1995). Optimal one and two-stage designs for the logistic regression model. Ph.D. dissertation, Virginia Polytechnic and State University, Virginia.
- MEYER, R.D., STEINBERGH, D.M & BOX, G.E.P. (1996). Follow-up designs to resolve confounding in fractional factorials. *Technometrics*, **38**, 303–313.
- MYERS, R.H. & MONTGOMERY, D.C. (2002). *Response Surface Methodology: Process and Product Improvement with Designed Experiments*, New York, John Wiley & Sons.
- NEFF, A. (1996). Bayesian two-stage designs under model uncertainty, Ph.D. dissertation, Virginia Polytechnic and State University, Virginia.
- NEFF, A. & MYERS, R.H. (1998). Recent developments in response surface methodology and its application in industry, in S. Park and G.G. Vining (eds.), *Statistical Process Control for Quality and Productivity Improvement*, New York: Marcel Dekker.
- RUGGOO, A. & VANDEBROEK, M. (2004). Bayesian sequential D-D optimal model-robust designs, *Computational Statistics and Data Analysis*, **47**(4): 655–673.
- WIENS, D. (1994). Robust designs for approximately linear regression: M-estimated parameters. *Journal of Statistical Planning and Inference*, **40**, 135–160.
- YUE, R-X. & HICKERNELL, F.J. (1999). Robust designs for fitting linear models with misspecification. *Statistica Sinica*, **9**, 1053–1069.